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10 / 10 submitted

Q1. Frequency Count

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    int count[100] = {0};
    int x, ch;
    int first = 1;

    while (scanf("%d",&x) == 1)
    {
        count[x]++;

        ch = getchar();

        if (ch == '\n' || ch == EOF)
            break;
    }

    for (x = 0; x < 100; x++)
    {
        if (count[x] > 0)
        {
            if (!first)
                printf(",");

            printf("%d:%d", x, count[x]);
            first = 0;
        }
    }

    return 0;
}
```

INPUT

1,1,2,2,3,3,3

OUTPUT

1:2,2:2,3:3

Q2. Palindrome String

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    char s[100];
    int n = 0;
    int i;
    int ok = 1;

    while (1)
    {
        s[n] = getchar();

        if (s[n] == '\n' || s[n] == EOF)
            break;

        n++;
    }

    for (i = 0; i < n / 2; i++)
    {
        if (s[i] != s[n - i - 1])
        {
            ok = 0;
            break;
        }
    }

    if (ok == 1)
        printf("Palindrome");
    else
        printf("Not palindrome");

    return 0;
}
```

INPUT

madam

OUTPUT

Palindrome

Q3. Missing Number

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int a[100];
    int n = 0, i;
    int sum = 0;

    while (scanf("%d", &a[n]) == 1) {
        n++;
    }

    for (i = 1; i ≤ n + 1; i++) {
        sum += i;
    }

    for (i = 0; i < n; i++) {
        sum -= a[i];
    }

    printf("%d", sum);

    return 0;
}
```

INPUT

1 2 3 5

OUTPUT

4

Q4. Valid Anagram

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char a[100], b[100];
    int count[256] = {0};
    int i;

    scanf("%99[^,],%99s", a, b);

    for (i = 0; a[i] != '\0'; i++)
        count[(unsigned char)a[i]]++;

    for (i = 0; b[i] != '\0'; i++)
        count[(unsigned char)b[i]]--;

    for (i = 0; i < 256; i++) {
        if (count[i] != 0) {
            printf("Not anagram");
            return 0;
        }
    }

    printf("Anagram");

    return 0;
}
```

INPUT

listen,silent

OUTPUT

Anagram

Q5. Merge Strings Unique

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    char s[200];
    int used[256] = {0};
    int i;

    scanf("%199s", s);

    for (i = 0; s[i] != '\0'; i++) {

        if (s[i] == ',')
            continue;

        if (used[(unsigned char)s[i]] == 0) {
            printf("%c", s[i]);
            used[(unsigned char)s[i]] = 1;
        }
    }

    return 0;
}
```

INPUT

abc,bcd

OUTPUT

abcd

Q6. Merge Overlapping Intervals

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    int a[100], n = 0;
    int x, ch;
    int i, j, temp;

    while (scanf("%d", &x) == 1)
    {
        a[n++] = x;
        ch = getchar();

        if (ch == '\n' || ch == EOF)
            break;
    }

    /* Sort the intervals by starting value */
    for (i = 0; i < n - 2; i += 2)
    {
        for (j = i + 2; j < n; j += 2)
        {
            if (a[i] > a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;

                temp = a[i + 1];
                a[i + 1] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }

    /* Merge overlapping intervals */
    i = 0;

    while (i < n)
    {
        int start = a[i];
        int end = a[i + 1];
```

```
j = i + 2;

while (j < n && a[j] ≤ end)
{
    if (a[j + 1] > end)
        end = a[j + 1];

    j += 2;
}

printf("%d,%d", start, end);

if (j < n)
    printf(",");

i = j;
}

return 0;
}
```

INPUT

1 4,4 5

OUTPUT

1,5

Q7. Reverse Words

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    char s[200];
    int n = 0;
    int i, j;

    /* Read the sentence */
    while (1)
    {
        char ch = getchar();

        if (ch == '\n' || ch == EOF)
            break;

        s[n] = ch;
        n++;
    }

    /* Reverse the order of words */
    i = n - 1;

    while (i ≥ 0)
    {
        /* Ignore spaces */
        while (i ≥ 0 && s[i] == ' ')
            i--;

        if (i < 0)
            break;

        j = i;

        /* Find start of current word */
        while (j ≥ 0 && s[j] ≠ ' ')
            j--;

        /* Print the word */
        for (int k = j + 1; k ≤ i; k++)
            putchar(s[k]);

        if (j ≥ 0)
```

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```
    putchar(' ');

    i = j - 1;
}

return 0;
}
```

INPUT

Data Structures and Algorithms

OUTPUT

Algorithms and Structures Data

Q8. First Non-Repeating Character

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    char s[200];
    int count[256] = {0};
    int i = 0;

    while (1)
    {
        char ch = getchar();

        if (ch == '\n' || ch == EOF)
            break;

        s[i] = ch;
        count[(unsigned char)ch]++;
        i++;
    }

    for (int j = 0; j < i; j++)
    {
        if (count[(unsigned char)s[j]] == 1)
        {
            printf("%c", s[j]);
            return 0;
        }
    }

    printf("No unique character");

    return 0;
}
```

INPUT

aabbcbcd

OUTPUT

d

Q9. Maximum Subarray

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    int a[100];
    int i = 0;
    int current, maximum;

    while (scanf("%d", &a[i]) == 1)
    {
        i++;
    }

    current = a[0];
    maximum = a[0];

    for (int j = 1; j < i; j++)
    {
        if (current + a[j] > a[j])
            current = current + a[j];
        else
            current = a[j];

        if (current > maximum)
            maximum = current;
    }

    printf("%d", maximum);

    return 0;
}
```

INPUT

-5 -2 -8 -1

OUTPUT

-1

Q10. Sort Characters by Frequency

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    char s[100];
    int freq[256] = {0};
    int i, j, n = 0;
    int ch;

    while ((ch = getchar()) != '\n' && ch != EOF)
    {
        s[n] = ch;
        freq[ch]++;
        n++;
    }

    /* Print characters according to frequency */
    for (i = n; i > 0; i--)
    {
        for (j = 0; j < 256; j++)
        {
            if (freq[j] == i)
            {
                int k;

                for (k = 0; k < i; k++)
                {
                    printf("%c", j);
                }

                freq[j] = 0;
            }
        }
    }

    return 0;
}
```

INPUT

cccaaa

OUTPUT

aaaccc